

AMMAR YASSER ABDALLAH

AI AND MACHINE LEARNING ENGINEER

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PROFESSIONAL SUMMARY

AI and Machine Learning Engineer working with Python, scikit-learn, retrieval-augmented generation (RAG), and computer vision. Builds reproducible model pipelines and serves AI functionality through FastAPI and interactive applications. Graduation project MindGuard combines a bilingual chatbot and wearable signal analysis with patient support workflows. Other projects cover classification, regression, realtime vision, and robotics.

CORE SKILLS

Languages and Data Analysis: Python, C++, Pandas, NumPy, Matplotlib, Jupyter
Machine Learning: scikit-learn, classification, regression, cross-validation, GridSearchCV, feature analysis, model evaluation
Generative AI and NLP: RAG, LangChain, LangGraph, Gemini API, BM25, hybrid retrieval, re-ranking
Databases and Infrastructure: SQL, ChromaDB, MongoDB, Docker
Computer Vision and Signals: OpenCV, MediaPipe, YOLOv8, feature extraction, HRV and PPG analysis
Applications and Robotics: FastAPI, Streamlit, REST APIs, NestJS, Socket.IO, ESP32, Arduino
Validation and Diagnostics: pytest, holdout evaluation, leakage prevention, residual analysis, feature importance
Automation and Collaboration: n8n, custom GPTs, LLM APIs, Git, GitHub, technical instruction, mentoring

PROFESSIONAL EXPERIENCE

Freelance AI and Full Stack Developer 2025 - Present
Self-Employed | Remote

- Built Python classification and regression pipelines, comparing models through cross-validation and separate holdout evaluation, with attention to data leakage.
- Developed realtime computer vision tools for landmarks, gesture recognition, and object detection, including a separate YOLOv8 fine-tuning workflow.
- Created an ESP32 biped control system with firmware APIs, Python interfaces, computer vision tracking, and calibrated movement sequences.

AI and Machine Learning Trainee Jun 2026 - Aug 2026
Infolith | Remote

- Completed structured AI and machine learning projects, practical assignments, and required assessments through the Infolith training portfolio.

Coding Instructor Feb 2025 - Mar 2026
O-Academy | Remote

- Taught live, project-based lessons in Python, AI foundations, Scratch, and application logic to U.S.-based school-age students.
- Adapted technical concepts into accessible exercises and mentored students with different levels of experience.

Full Stack Developer Intern Sep 2025 - Nov 2025
MAIM Digital Solutions | Alexandria, Egypt

- Built MongoDB data models, Node.js and Express REST APIs, and responsive React interfaces during the MAIM Web Summer Training program.
- Integrated APIs, resolved defects, and developed reusable components. Reviewed code and contributed pull requests within an Agile Git workflow.

AI and Automation Engineer Apr 2025 - Sep 2025
Armstrong | Alexandria, Egypt

- Automated educational content workflows for the Digital Egypt Cubs Initiative using n8n, custom GPTs, LLM APIs, and open-source models.
- Translated educational requirements into reusable prompts and orchestration steps, validated outputs, and troubleshoot integrations.

SELECTED PROJECTS

MindGuard | Graduation Project in Mental Health Support

Python, FastAPI, NestJS, LangGraph, LangChain, Gemini, ChromaDB, BM25, scikit-learn

- Built a safety-aware bilingual chatbot using LangGraph orchestration, Gemini, and retrieval components for mental health support, with interactive testing through Streamlit.
- Developed FastAPI services for HRV-based stress analysis, sleep prediction, GAD-7 screening, and correlation analysis, using wearable signal processing and holdout or out-of-fold model evaluation.
- Connected AI functionality to the patient web experience through NestJS application services and Socket.IO messaging, with streamed NOVA responses and persistent conversation history.

VisionDeck | Realtime Computer Vision Suite

Python, OpenCV, MediaPipe, YOLOv8, Streamlit

- Combined face landmark tracking, hand gesture recognition, and object detection in an interactive computer vision application.
- Supported webcam input and automatic IP-camera discovery, and developed a separate YOLOv8 fine-tuning workflow for custom object classes.

Biped Robot Control System | Robotics and Vision Platform

Python, C++, ESP32, OpenCV, Streamlit, HTTP APIs

- Built an ESP32 biped with persistent pose storage and firmware APIs, controlled through desktop and Streamlit interfaces.
- Implemented automatic device discovery, color-based robot and ball tracking, and a calibrated autonomous movement sequence for forward motion and stopping.

Tune a Classifier to Beat a Baseline | Reproducible Classification Experiment

Python, scikit-learn, Pandas, NumPy, GridSearchCV, pytest

- Created a leakage-resistant classification pipeline with a separate holdout set, cross-validation, and hyperparameter search across candidate classifiers.
- Compared tuned models against a baseline using ROC-AUC and included automated tests to check the pipeline.

House Price Predictor | Regression Pipeline

Python, scikit-learn, Pandas, NumPy, Matplotlib, pytest

- Compared regression algorithms through cross-validation and evaluated a Gradient Boosting model on a separate test set.
- Added feature-importance analysis, residual diagnostics, and pipeline tests to examine model behavior and validate the regression workflow.

EDUCATION

Bachelor's Degree in Computers and Data Science

2026

Alexandria University

Relevant Coursework: Machine Learning, Natural Language Processing, Pattern Recognition, Neural Networks